Beyond light signals and single linear features:

The stimulus, s, need not be a spatiotemporal pattern of light.

Ex: For the H1 neuron in the blowfly,

$$s(t) = v(t)$$

is the velocity of image motion in typical models. The firing rate is then a temporal convolution followed by nonlinearity:

$$r = f((K * s)(t)).$$

This neuron is "wide field," rather than having a narrow receptive field. More accurate models permit multiple linear features:

$$r = \underbrace{f}_{\substack{n\text{D} \\ \text{nonlinearity}}}\Big(\underbrace{(K_1 * s)(t)}_{\text{feature } 1}, \underbrace{(K_2 * s)(t)}_{\text{feature } 2}, \cdots, \underbrace{(K_n * s)(t)}_{\text{feature } n}\Big)$$

For H1, 4 significant temporal features have been identified.

Compute additional features through "spike-triggered covariance" (STC). Features are eigenvectors.

$$C_{p_1 p_2} = \frac{1}{n_s} \sum_{i=1}^{n_s} \left(s_{p_1 t_i} - \bar{s}_{p_1}\right)\left(s_{p_2 t_i} - \bar{s}_{p_2}\right)$$

When there are stimulus correlations (i.e.: it's not perfectly white), instead compute eigenvectors of $C - \underbrace{C^{(x)}}_{\substack{\text{stimulus} \\ \text{correlations}}}$. Like STA, f is estimated by plotting

$$P_{spike}(\hat{L}_1, \hat{L}_2, \cdots, \hat{L}_n).$$

Generalized linear models:

Three elements of GLM, canonical form

1. Probabilistic model of data, $\{y_\mu\}_{\mu=1}^{P}$

   Exponential family:

$$P(y|\theta) = \underbrace{\prod_{\mu=1}^{P}}_{\text{independent samples}} \exp\Big(y_\mu \underbrace{\theta_\mu}_{\text{"canonical parameter"}} + \underbrace{H(y_\mu) - D(\theta_\mu)}_{\text{arbitrary functions}}\Big)$$

   Examples: Normal distribution, Poisson, Bernoulli

2. A linear predictor of canonical parameter

$$\theta_\mu = \sum_{n=1}^{N} \beta_n x_{n\mu} \quad , \quad \begin{array}{l}\beta_n : \text{weighting coefficients} \\ x_{n\mu} : \text{independent variables}\end{array}$$

3. A "link function" relating the distribution's mean and canonical parameter

$$\theta_\mu = \underbrace{g}_{\text{link function}}(\underbrace{m_\mu}_{\text{mean}}) \quad , \quad m_\mu = g^{-1}(\theta_\mu)$$

Given a GLM, the weighting coefficients, β, are chosen to maximize the likelihood of the data.

$$\hat{\beta} = \arg\max_{\beta} \ell(\beta) = \arg\max_{\beta} \log P(y|\theta(\beta))$$

In practice, it's easy to ~~optimi~~ optimize GLMs in canonical form numerically. Very popular method in neuro!

Unit normal GLM: ordinary linear regression

1. $$P(y|m) = \prod_{\mu=1}^{P} \frac{1}{\sqrt{2\pi}} e^{-(y_\mu - m_\mu)^2/2}$$

$$= \prod_{\mu=1}^{P} \exp\Big(\underbrace{y_\mu m_\mu}_{\theta_\mu} \underbrace{- \tfrac{1}{2} y_\mu^2 - \log\sqrt{2\pi}}_{H(y_\mu)} \underbrace{- \tfrac{1}{2} m_\mu^2}_{-D(\theta_\mu)} \Big)$$

The canonical parameter is the mean.

2. $$\theta_\mu = \sum_{n=1}^{N} \beta_n X_{n\mu}$$

3. The canonical link function is the identity

$$g(x) = x \Rightarrow \theta_\mu = m_\mu$$

$$m_\mu = \sum_{n=1}^{N} \beta_n X_{n\mu}$$

The resulting log-likelihood function is

$$\ell(\beta) = -\underbrace{\sum_{\mu=1}^{P} \Big(\tfrac{1}{2}\big(y_\mu - \sum_{n=1}^{N} \beta_n X_{n\mu}\big)^2}_{\text{Proportional to mean squared error}} + \underbrace{\log\sqrt{2\pi}}_{\text{independent of } \beta} \Big)$$

Maximizing the log-likelihood is equivalent to minimizing the mean squared error.

Poisson GLM: Poisson regression

1. $$P(y|\lambda) = \prod_{\mu=1}^{P} \frac{\lambda_\mu^{y_\mu} e^{-\lambda_\mu}}{y_\mu!}$$

$$= \prod_{\mu=1}^{P} \exp\Big(y_\mu \underbrace{\log\lambda_\mu}_{\theta_\mu = \log\lambda_\mu} - \log y_\mu! - \lambda_\mu \Big)$$

$$P(y|\theta) = \prod_{\mu=1}^{P} \exp\Big(y_\mu \theta_\mu \underbrace{- \log y_\mu!}_{H(y_\mu)} \underbrace{- e^{\theta_\mu}}_{-D(\theta_\mu)} \Big)$$

2. $$\theta_\mu = \sum_{n=1}^{N} \beta_n X_{n\mu}$$

3. Since the mean is λ_μ, and $\theta_\mu = \log \lambda_\mu$, the canonical link function is

$$g(x) = \log(x) \Rightarrow \lambda_\mu = e^{\theta_\mu}$$

$$\lambda_\mu = e^{\sum_n \beta_n x_{n\mu}}$$

The resulting log-likelihood function is

$$\ell(\beta) = \sum_{\mu=1}^{P} \Big(y_\mu \sum_{n=1}^{N} \beta_n x_{n\mu} - e^{\sum_n \beta_n x_{n\mu}} \underbrace{- \log y_\mu!}_{\text{independent of } \beta} \Big)$$

GLM model of RGCs:

In 2008, Jonathan Pillow and Eero Simoncelli popularized GLM modeling in neuroscience by fitting spiking data from many RGCs with a neural network instantiation of the Poisson GLM.

Comparison of GLM to LNP model:

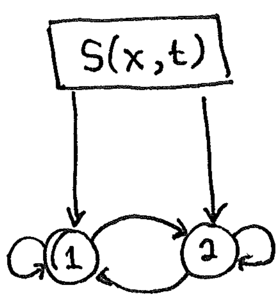
Take $\mu = t$,

$$k_t \sim \underbrace{\text{Poiss}}_{P}\Big(\underbrace{\exp}_{N}\big(\underbrace{\textstyle\sum_{n=1}^{N} \beta_n X_{nt}}_{L}\big)\Big)$$

Rather than X being white noise stimuli, X can be anything. Rather than fitting N, it's defined as $g^{-1}(x) = e^{x}$. Note that N is 1D.

Pillow model:

X comprised: binary white noise stimuli, past spikes of neuron, past spikes of other neurons.



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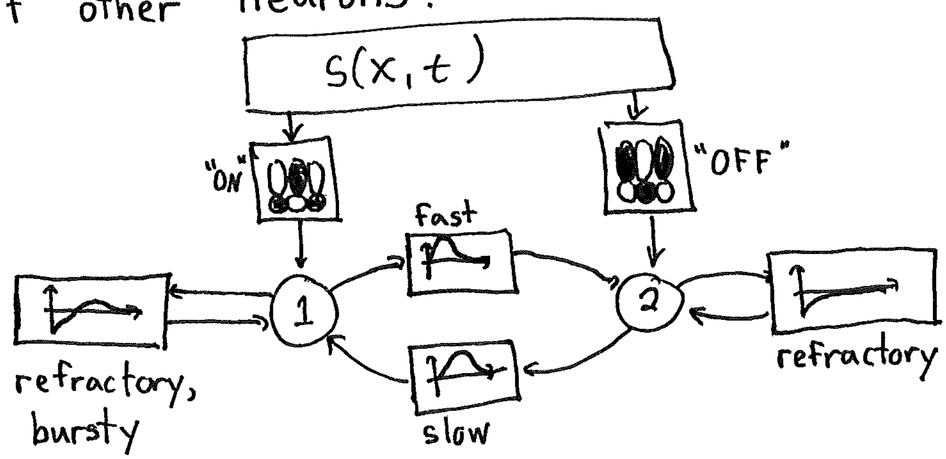

All these linear filters fit at once directly to data.
Actual model had 27 RGCs, not 2.

Bernoulli GLM: logistic regression

1. $P(y|q) = \prod_{\mu=1}^{P} q_\mu^{y_\mu}(1-q_\mu)^{1-y_\mu}$, $y_\mu \in \{0,1\}$, $q_\mu = P(y_\mu = 1)$

$$\theta_\mu = \log \frac{q_\mu}{1-q_\mu} \quad , \text{"log odds ratio"}$$

$$P(y|\theta) = \prod_{\mu=1}^{P} \exp\Big(y_\mu \theta_\mu \underbrace{- \log(1+e^{\theta_\mu})}_{-D(\theta_\mu)}\Big)$$

2. $\theta_\mu = \sum_{n=1}^{N} \beta_n x_{n\mu}$

3. Since the mean is q_μ

$$\theta_\mu = \log \frac{q_\mu}{1-q_\mu} \Rightarrow g(x) = \log \frac{x}{1-x} \Rightarrow q_\mu = \frac{e^{\theta_\mu}}{1+e^{\theta_\mu}}$$

$$q_\mu = \frac{e^{\sum_n \beta_n x_{n\mu}}}{1+e^{\sum_n \beta_n x_{n\mu}}} \quad , \quad 1 - q_\mu = \frac{1}{1+e^{\sum_n \beta_n x_{n\mu}}}$$

The resulting log-likelihood function is

$$\ell(\beta) = \underbrace{\sum_{\mu=1}^{P} \Big(y_\mu \log q_\mu(\beta) + (1-y_\mu)\log(1-q_\mu(\beta))\Big)}_{\text{Negative of cross-entropy loss mentioned in machine learning lecture (classification)}}$$

Neuroscience applications common when studying binary behaviors (e.g.: choose A or B; Go or No/Go; Attack or not; etc.)